

SALARY PREDICTION

Comparing Ensemble Regression Models to Predict Individual Salary

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IBM Project • PBEL 3.0 AI

Project Objective

Predict an individual's salary from demographic and professional attributes, then compare four ensemble learning models to identify which one generalizes best.

Age

Gender

Education Level

Job Title

Years of Experience

6,704 records

Target Variable

Salary (continuous, INR)

Predicted via regression using four ensemble methods, evaluated on a held-out 20% test split.

Data Preparation & Cleaning

01

Handle Missing Values

Rows with null entries were dropped to ensure clean, reliable inputs for modeling.

02

Consolidate Job Titles

Job titles appearing 25 times or fewer were grouped into an 'Others' category to reduce noise from rare categories.

03

Normalize Education Levels

Inconsistent labels (e.g. "Bachelor's Degree", "pHD") were standardized to consistent categories.

04

One-Hot Encoding

All categorical variables (Gender, Education Level, Job Title) were converted to numeric dummy variables.

Modeling Approach

Four ensemble regressors were trained on an 80/20 train-test split (random_state = 42).

Random Forest

Bagged decision trees; averages predictions to reduce variance.

Gradient Boosting

Sequentially fits trees to residual errors, boosting accuracy.

AdaBoost

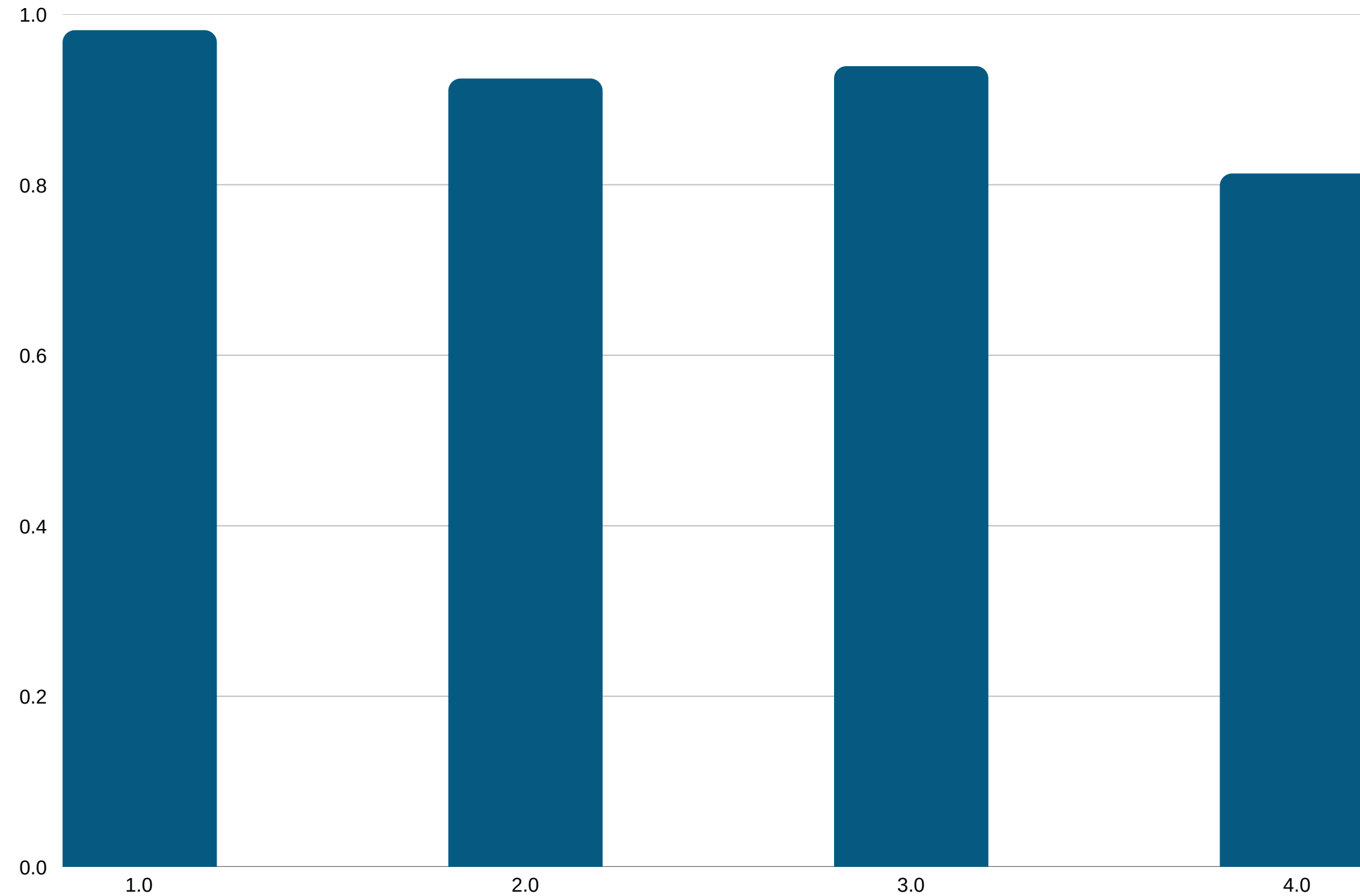
Reweights misclassified points each round to focus learning.

Voting Regressor

Combines RF, GB & AdaBoost predictions into one ensemble.

Model Comparison — R^2 Score

Proportion of variance in salary explained by each model (higher is better)



BEST MODEL

Random Forest

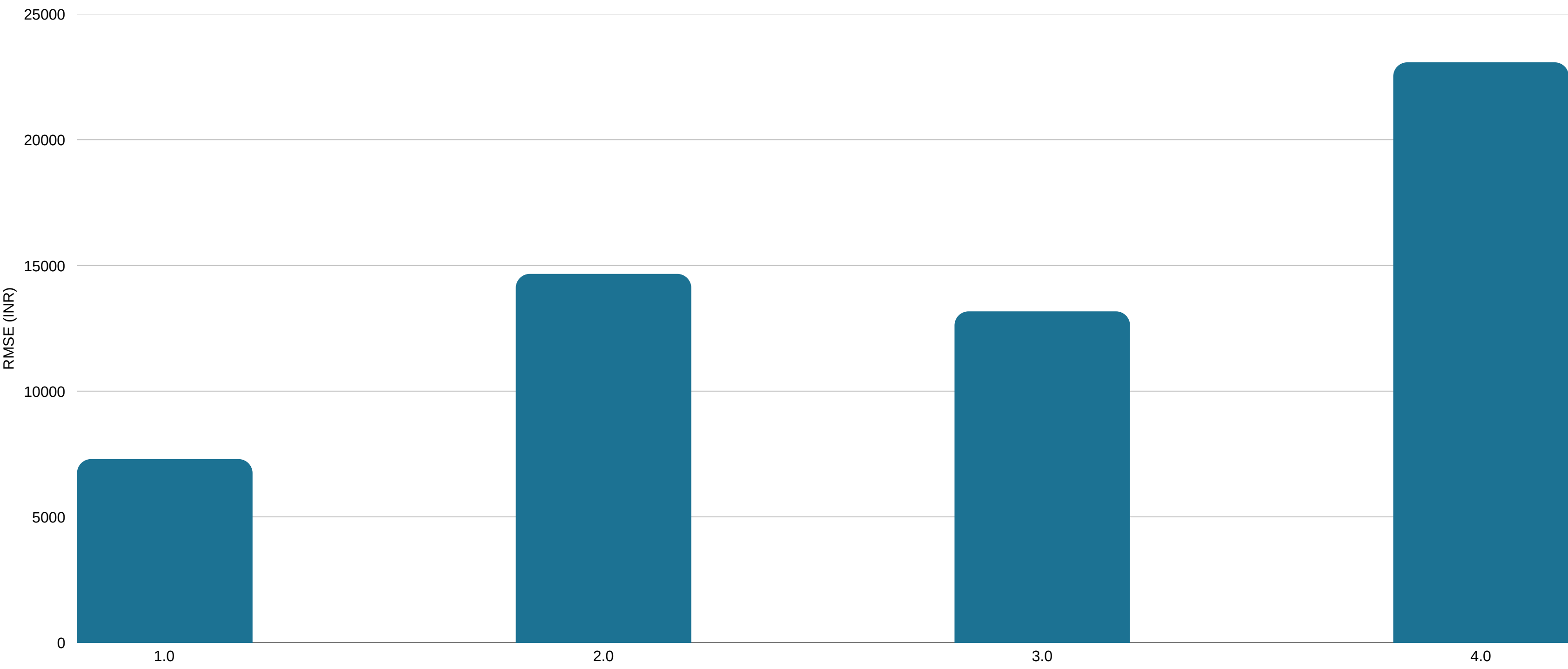
0.9813

R^2 Score on held-out test data

Explains about 98.1% of the variance in salary
— the strongest fit of the four models tested.

Model Comparison — Error Metrics

Root Mean Squared Error, in rupees (lower is better)



Full Results Table

Model	R ² Score	MSE	RMSE (\$)
Random Forest	0.9813	53,254,937	7,298
Gradient Boosting	0.9246	215,008,153	14,663
AdaBoost	0.8132	532,515,686	23,076
Voting Regressor	0.9391	173,519,622	13,173

Random Forest wins on every metric

It posts the highest R² and the lowest MSE/RMSE simultaneously — a clean, unambiguous winner rather than a trade-off between metrics.

Random Forest: Actual vs. Predicted Salary

Predictions cluster tightly around the ideal reference line, confirming the model captures the underlying salary patterns well.

$R^2 = 0.9813$

Most points fall within a tight band of the $y = x$ reference line, with only a small number of higher-salary outliers deviating — consistent with the model's high R^2 and low RMSE.

See the notebook's scatter plot (Actual vs Predicted Salary) for the full visualization.

7,298

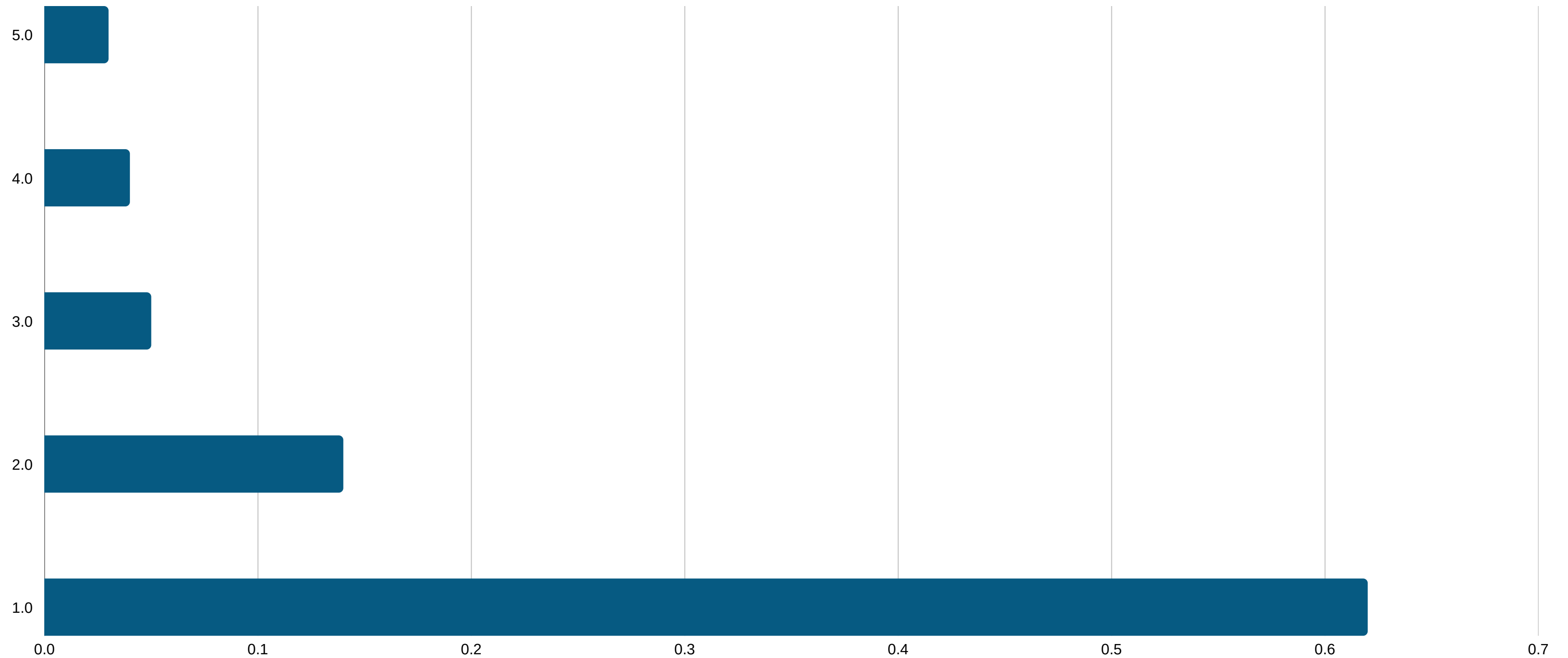
RMSE (USD) — average prediction error

Why this matters

A tight actual-vs-predicted spread means the model can be trusted for realistic salary benchmarking, not just headline accuracy metrics.

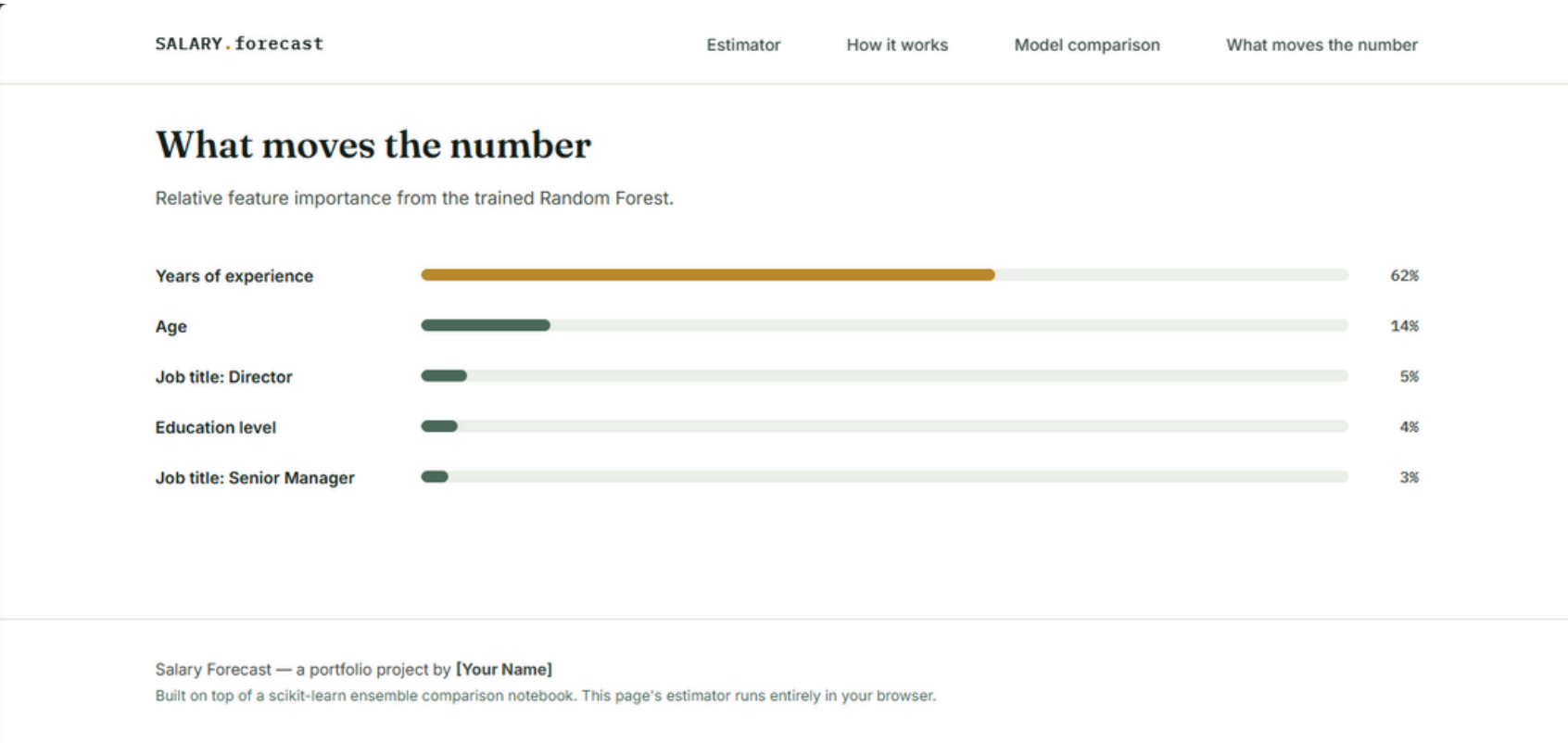
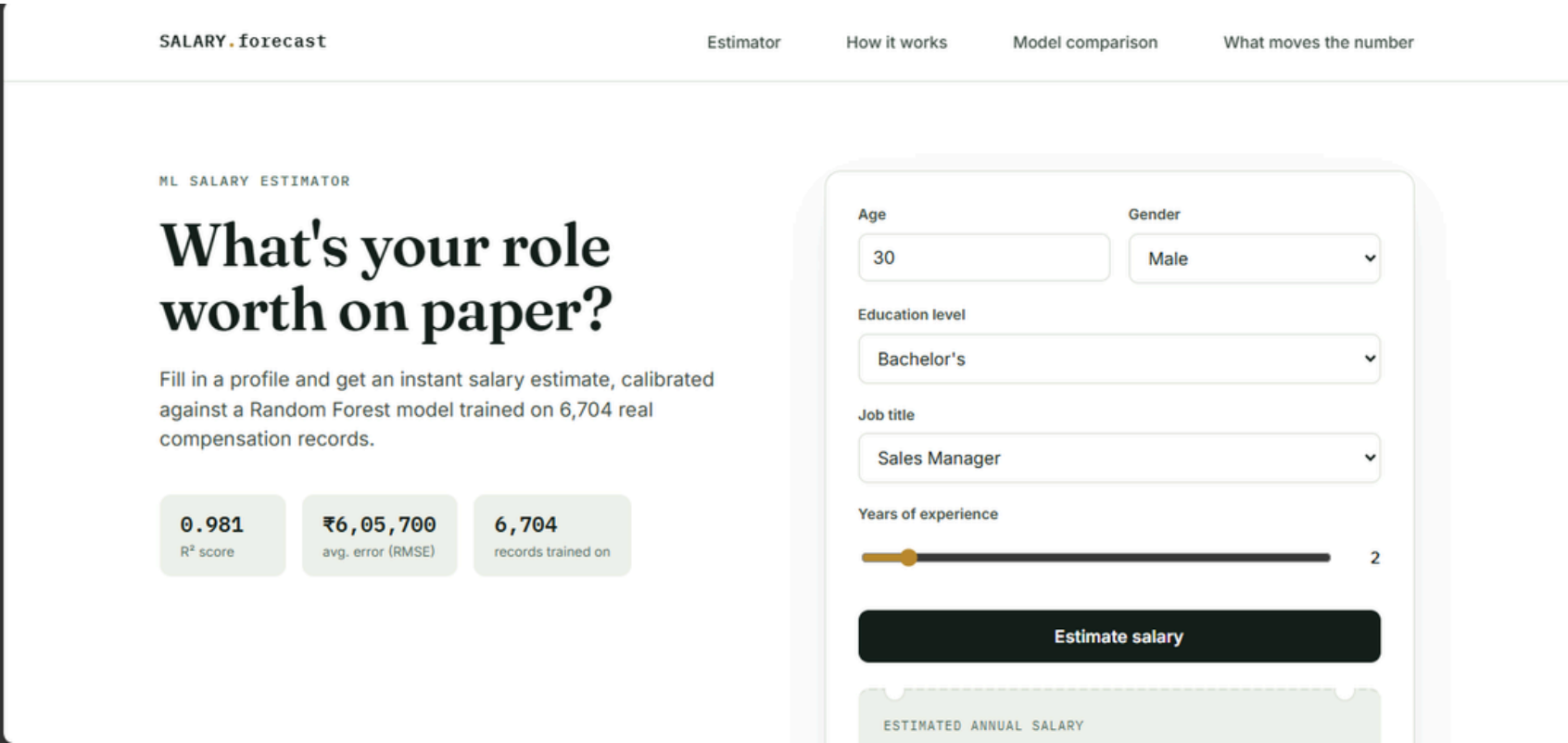
What Drives Salary the Most?

Top feature importances from the Random Forest model



Deployed as a Web Application

A client-side salary estimator, styled and calibrated to the same feature-importance pattern the trained model learned



Estimator page

Instant salary estimate from age, education, job title and years of experience.

What moves the number

Mirrors the trained Random Forest: experience dominates, then age and job title.

Built with HTML, CSS & JavaScript • Runs entirely client-side • Static heuristic, not a live model call

Key Takeaways

- Random Forest was the clear winner — highest R^2 (0.9813) and lowest RMSE (₹ 7,298) among all four ensemble models tested.
- Years of Experience dominates salary prediction, far outweighing Age, Job Title, or Education Level.
- Ensemble methods that average many trees (Random Forest, Voting) outperformed boosting-only approaches on this dataset.
- The model explains ~98% of salary variance, making it reliable for practical compensation benchmarking.
- Also shipped as a lightweight, client-side web app that mirrors the model's feature-importance ranking.

Thank you